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| **Fiscal Year:** | FY 2010 | **Task Last Updated:** | FY 11/02/2009 |
| **PI Name:** | Azzam, Edouard Ph.D. | | |
| **Project Title:** | The Role of Gap-Junction Communication and Oxidative Metabolism in the Biological Effects of Space Radiation | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | azzamei@njms.rutgers.edu | **Fax:** | FY (973) 972-1865 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | (973) 972-5323 |
| **Organization Name:** | RUTGERS Biomedical and Health Sciences - New Jersey Medical School | | |
| **PI Address 1:** | New Jersey Medical School Cancer Center - Department of Radiology | | |
| **PI Address 2:** | 205 S Orange Ave, Cancer center - F1012 | | |
| **PI Web Page:** | | | |
| **City:** | Newark | **State:** | NJ |
| **Zip Code:** | 07103 | **Congressional District:** | 10 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2006 Radiation Biology NNJ06ZSA001N |
| **Start Date:** | 01/01/2007 | **End Date:** | 12/31/2011 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Cucinott1a, Francis | **Contact Phone:** | 281-483-0968 |
| **Contact Email:** | noaccess@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date extended until 12/31/2011, per C. Guidry/JSC (10/2010) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | de Toledo , Sonia ( UMDNJ-New Jersey Medical School )<br>Harris, Andrew ( New Jersey Medical School (UMDNJ) )<br>Howell, Roger ( University of Medicine and Dentistry of New Jersey )<br>Pain, Debkumar ( UMDNJ - New Jersey Medical School ) | | |
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| **Performance Goal Text:** | | | |

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| **Task Description:** | The objective of this project is to investigate biochemical and molecular aspects of two inter-related fundamental biological processes, namely gap-junction intercellular communication and oxidative metabolism, in determining human cellular responses to exposures of very low to moderate fluences of protons (of energies such as found in the space environment) or high charge (Z) and high energy (E) particles (1 GeV/nucleon 56Fe ions). Gap junctions are found between most cells in the body, and the molecular signaling they mediate is crucial for normal development, physiology and response to disease. By allowing direct intercellular transfer of cytoplasmic molecules, they provide a powerful pathway for direct molecular signaling between cells. On the other hand, oxidative metabolism is primarily regulated by mitochondria, which mediate respiration and ATP synthesis in eukaryotic cells. They also participate in numerous indispensable metabolic pathways. We hypothesize that exposure of human cells to high or low fluences of protons and/or HZE particles results in distinct molecular and biochemical changes not only in irradiated but also neighboring bystander cells. These events are modulated by oxidative metabolism and gap-junction coupling, which together induce persistent changes in mitochondria and in nuclear DNA in progeny of irradiated and affected bystander cells. In Aim 1, we propose to define the aspect(s) of junctional communication leading to proton- and high charge/high energy (HZE) particle-induced toxic effects. Specifically, we will examine the role of different connexin proteins in propagating death-inducing (growth delay, apoptosis) or pro-survival effects (proliferative capacity) among irradiated cells and also between irradiated and non-irradiated cells. Aim 2 is to identify specific connexin channel permeabilities that correlate with intercellular propagation of toxic or protective effects in cultures exposed to HZE particles or protons. An emphasis will be to determine the nature of molecules communicated among irradiated cells and between irradiated and non-irradiated cells. In Aim 3, we will evaluate mitochondrial function in cells exposed to low fluences of high energy charged particles. Specifically, we will examine mitochondrial protein import, mitochondrial protein synthesis, assembly of large protein complexes including respiratory complexes, and oxidative damage to mitochondrial proteins and DNA. We will also evaluate the persistence of oxidative effects in the progeny of exposed cells and correlate the observed results with the overall cellular redox environment. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The proposed experiments relate to Risk No. 28, subsection 28b (molecular studies on carcinogenesis) in NASA's 'Roadmap'. In addition to revealing signaling pathways underlying non-targeted effects of ionizing radiation, the outcome of this proposal may provide insight into space radiation-induced degenerative diseases such as cognitive defects and cardiovascular disorders (Risks No. 29 and 30). It will provide a mechanistic basis for understanding exposures and risks to human from low fluences of high and low linear energy transfer radiations. By identifying the nature of molecules communicated among irradiated and neighboring non-irradiated cells, it may contribute to remediation strategies against harmful effects of space radiation, and is also relevant to heavy ion and proton radiotherapy. |

In repeat experiments, we have generated strong evidence for the involvement of gap junction communication and oxidative metabolism in the propagation of stressful effects from irradiated normal human cells to their neighboring non-irradiated bystander cells. Importantly, we have shown that the linear energy transfer (LET) of the radiation is critical in the expression of the stressful bystander effects. Further, high LET-induced stressful effects in bystander cells are transmitted to their progeny.
Whereas the experiments during the past year have confirmed earlier findings, they have also revealed novel molecular and biochemical events implicated in the role of gap-junction permeability and oxidative metabolism in the cellular responses to space radiation, and in the persistence of stressful effects in the progeny of irradiated cells and their neighboring bystanders.

I. The role of intercellular communication:

a) Using a layered tissue culture strategy that allows separation of irradiated and bystander cells, we have shown that stressful effects are propagated from energetic iron ion (1 GeV/amu)-exposed normal human cells to their neighboring non-irradiated bystander cells. Micronuclei, a surrogate form of DNA damage, were induced in bystander cells incubated with cells from cultures exposed to mean doses as low as 10 cGy. Gap junction intercellular communication is the major mediator of the induced bystander effect detected. Incubation of the cell cultures with gap junction inhibitors prevented the propagation of stressful bystander effects, including effects leading to DNA damage, oxidative stress and up-regulation of the stress-inducible p53 signaling pathway. In contrast, secreted diffusible factors or incubation with lanthanum, an inhibitor of hemi-channels, had no significant effect.

b) Further support for the expression of bystander effects and the mediating role of gap junctions was generated in monolayer cultures of quiescent normal human fibroblasts exposed to heavy ions. The p53 signaling pathway was up-regulated following exposure to mean doses as low as 0.2 cGy from 1 GeV/amu iron ions, by which ~ 1% of the cells are traversed through the nucleus by a particle track. The effect correlated with a greater increase in 53BP1 foci (a marker of DNA damage) than expected, based on the number of cells traversed by a heavy ion track. When the cell cultures were irradiated in the presence of gap junction inhibitors, the increase in the level of the stress response did not occur. Ongoing experiments with culture dishes that integrate an ultra-thin solid state nuclear track detector, to identify with micron precision the irradiated cells, will measure the distance propagation of the bystander effect.

c) Consistent with the above results, exposure of confluent cell cultures to very low fluences of energetic iron ions resulted in up-regulation of connexin 43, a constitutive protein of gap junctions. Increased phosphorylation, and sequestering of connexin 43 in lipid rafts of the plasma membrane, correlated with functional junctional communication among irradiated cells and between irradiated and non-irradiated bystander cells.

d) Extensive data have shown that the selective permeability of the gap-junction channel linking irradiated and bystander cells with each other determines the nature of the propagated bystander effect. Whereas junctional channels composed of connexin 26 mediated the propagation of stressful bystander effects, gap junctions composed of connexin 32 protected cells against the lethal and clastogenic effects of space radiation.

e) We have compared cell killing and DNA damage induced by energetic protons or iron ions (1GeV/amu) in normal human fibroblasts (AG1522 cells), and contrasted the results with data obtained in AG1522 cells exposed, in parallel, to cesium-137 gamma-rays or 3.5 MeV alpha-particles. As expected, a higher level of cell killing and DNA damage per unit dose was induced in cell cultures exposed to alpha-particles or energetic iron ions than energetic protons or gamma-rays. At 10% survival level, the relative biological effectiveness of energetic protons, alpha-particles and iron

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| **Task Progress:** | ions compared to gamma-rays were ~ 1.02, 5.8 and 3.2, respectively. Whereas incubating quiescent cells for several hours after exposure to gamma-rays or energetic protons increased survival, a decrease rather than an increase in survival occurred in alpha-particle- or iron ion-irradiated cells, respectively. The survival data mirrored results on induced DNA damage. Inhibiting junctional communication by down-regulating connexin 43 by siRNA, attenuated these stressful effects in alpha-particle- and iron ion-irradiated cells. Collectively, our data support the concept that cell populations or tissues respond as a whole to radiation exposure and the response is not restricted to that of the individual traversed cells but involves extensive communication among irradiated cells and between irradiated and bystander cells. They show that gap junction communication is a major mediator of the response of cell populations to high LET radiations.<br><br>II. The Role of Oxidative Metabolism:<br><br>Intracellular accumulation of oxidants results in modification of proteins, DNA and lipids. Persistence of such damages and their transmission to daughter cells may contribute to the development of cancer, atherosclerosis, accelerated aging and other degenerative diseases. In our studies, we have focused on oxidative stress resulting from space-radiation induced perturbation in oxidative metabolism, and examined changes in mitochondrial functions and modulation of antioxidant enzyme activity.<br><br>a) We have shown that low and high mean doses of 1GeV/amu iron ions disrupt mitochondrial physiology. Exposure of normal human cell cultures to mean doses as low as 3 cGy, resulted in similar impairment of mitochondrial protein import as exposure to 100 cGy. The decrease in import correlated with a decrease in mitochondrial membrane potential. Ongoing experiments are examining the modulation of expression of proteins that participate in the import process, namely the protein translocases in the outer and inner mitochondrial membranes.<br><br>b) We have also pursued our investigation of the modulation of aconitase activity in irradiated AG1522 cells. The iron-sulfur cluster in the active site of aconitase is essential for its catalytic activity, but it also renders the enzyme highly vulnerable to oxidative stress. Exposure of cell cultures to doses as low as 10 cGy from 1 GeV protons or 1 cGy from 1 GeV/amu iron ions down-regulated aconitase activity, but did not alter its protein level. The changes could be detected as late as 48 h after the exposure. Ongoing experiments are also examining the kinetics of modulation of the level/activity of antioxidant enzymes. Overall, the data support the strong connection between oxidative stress and the cellular responses to radiation exposure.<br><br>III. Persistence of Stressful Effects: Biological Responses in Progeny Cells:<br><br>a) Repeat experiments have indicated that stressful effects persist in the progeny of normal human cells exposed to low or high doses from 1GeV/amu iron ions. Elevated levels of micronuclei were observed in progeny cells up to 20 population doublings after exposure, which correlated with increases in mitochondrial superoxide anions and the amount of mitochondria. It also resulted in overall increase in cellular content of reactive oxygen species. In contrast, DNA damage and oxidative stress did not persist in progeny of proton (1 GeV/amu )-irradiated cells.<br><br>b) Of particular significance, oxidative stress (lipid peroxidation, protein carbonylation), DNA damage, and reduced cloning efficiency persisted also in progeny of bystander cells that were co-cultured with iron ion-irradiated cells. Significantly, the neoplastic transformation frequency in the progeny of bystander C3H/10T½ mouse embryo fibroblasts (MEFs) that were co-cultured with MEFs exposed to mean doses as low as 10 cGy from 1 GeV/amu iron ions was significantly elevated. When the bystander and irradiated cells were co-cultured in the presence of gap-junction inhibitors, the neoplastic transformation frequency in the progeny of bystander cells was similar to control.<br><br>IV. The role of LET in expression of adaptive responses to ionizing radiation:<br><br>Bystander cells that were co-cultured with cells exposed to low doses from 1 GeV protons were protected against DNA damage from a challenge dose of 50 cGy from 1 GeV/amu iron ions. The micronucleus frequency in the bystander cells was significantly lower than in respective control. The response is transient and is expressed during the first 24 h following isolation of the bystander cells from the co-culture. |
| **Bibliography Type:** | Description: (Last Updated: 07/29/2025) |
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| **Abstracts for Journals and Proceedings** | de Toledo SM, Buonanno M, Autsavapromporn N, Li M, Gonon G, Zhang J, Azzam EI. "The role of gap-junction communication and oxidative metabolism in the biological effects of space radiation." Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009.<br>Abstract Book 2009 July; page 224. Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009. , Jul-2009 |

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| **Abstracts for Journals and Proceedings** | Buonanno M, de Toledo SM, Howell RW, Pain D, Azzam EI. "Radiation Quality and the Induction of Long-Term Biological Effects in Irradiated Normal Human Cells and Neighboring Bystanders: the Role of Oxidative Metabolism." Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009.<br>Abstract Book 2009 July; page 129. Heavy Ions in Therapy and Space Symposium, Cologne, Germany, July 6-10, 2009. , Jul-2009 |
| **Abstracts for Journals and Proceedings** | de Toledo SM, Buonanno M, Autsavapromporn N, Li M, Gonon G, Zhang J, Pain D, Azzam EI. "The role of gap-junction communication and oxidative metabolism in the biological effects of space radiation." NASA Human Research Program Investigators' Workshop, Humans Return to the Moon, League City, TX, February 2-4, 2009.<br>Abstract Book 2009 Feb; abstract number 1078. NASA Human Research Program Investigators' Workshop, Humans Return to the Moon, League City, TX, February 2-4, 2009. , Feb-2009 |